MIL-N-45938/3B
4 March 1988
SUPERSEDING
MIL-N-45938/3A
28 JULY 1982

MILITARY SPECIFICATION SHEET

NUT, PLAIN, CLINCH AND NUT, SELF-LOCKING, CLINCH (SELF CLINCHING, KNURLED COLLAR, 450°F AND 600°F)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the nut described herein shall consist of this specification and the latest issue of MIL-N-45938

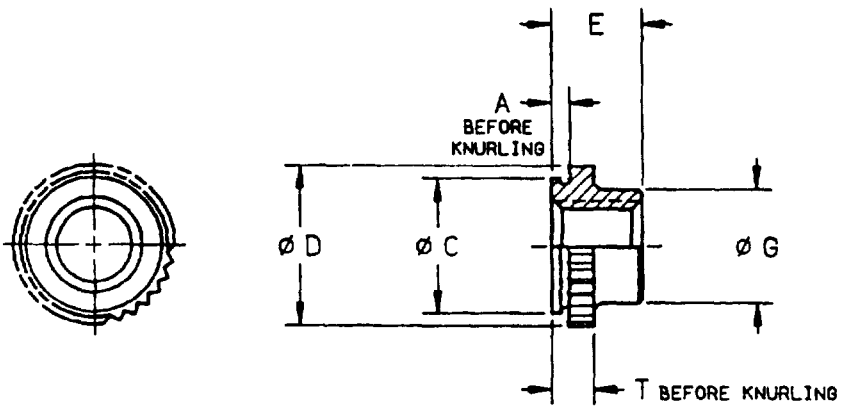


TABLE I Part dimensions and dash numbers

DASH NUMBER			THREAD CLASS 3B	A	SHEET THK MIN	HOLE SIZE IN SHEET	ØC	ØD	E	ØG	T	CENTER OF HOLE TO SHEET EDGE - MIN
ALLOY STEEL (450°F)	CRES 303 (450°F)	CRES A286 (600°F)	FIL-S-8879	± 004		+ 004 - 001	+ 001 - 004	± 015	± 015	± 005	± 005	SEE REQT 5
1	1C	---	0 0860-56 UNJC	030	040	201	199	260	178	125	078	188
2●	--	2C●				125	123	190	140	106		
3	3C	---	0 1120-40 UNJC	030	040	201	199	260	178	160	078	203
4●	--	4C●				156	154	220	160	135		
5	5C	---	0 1380-32 UNJC	030	040	228	226	290	178	175	078	250
6●	--	6C●				188	186	250	178	161		
7	7C	---	0 1640-32 UNJC	040	050	281	279	350	188	215	088	281
8●	--	8C●				219	217	280	188	200		
9	9C	---	0 1900-32 UNJF	040	050	281	279	350	188	240	088	281
10	10C	---	0 2500-28 UNJF	050	060	375	373	450	250	313	115	375
11	11C	---	0 3125-24 UNJF	063	080	406	404	490	312	386	141	406
12	12C	---	0 3750-24 UNJF	094	125	500	498	560	375	448	188	500
13	13C	---	0 1900-24 UNJC	040	050	281	279	350	188	240	088	281
14	14C	---	0 2500-20 UNJC	050	060	375	373	450	250	313	115	375
15	15C	--	0 3125-18 UNJC	063	080	406	404	490	312	386	141	406
16	16C	---	0 3750-16 UNJC	094	125	500	498	560	375	448	188	500

NOTES

1 Dimensions:
All dimensions are in inches

Ⓑ Denotes changes

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REQUIREMENTS:

1. Material:
Steel, alloy, for use in materials with a hardness of 28 HRC max
Steel, corrosion-resistant, type A286 (UNS S66286) per AMS5731 or AMS5734 (B)
for use in materials with a hardness of 90 HRB max
Steel, corrosion-resistant, type 303 (UNS S30300) per ASTM A581 or ASTM A582
for use in materials with a hardness of 90 HRB max

2 Protective Coating or Treatment:
(B) Steel, alloy, shall be cadmium plated per QQ-P-416 type III class 3, plus solid film lubricant coating
(B) Steel, corrosion-resistant, types 303 and A286, shall be cleaned and descaled in accordance with ASTM A380 and passivated in accordance with QQ-P-35
Self-locking, steel, corrosion resistant (type 303) nuts shall also have a solid film lubricant coating
Self-locking, steel, corrosion resistant (type A286) nuts shall be silver plated plus solid film lubricant coating.

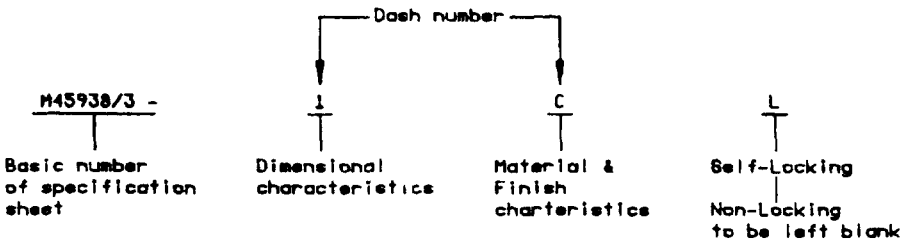
3 Threads:
The threads shall be in accordance with MIL-S-8879 prior to solid film lubricant coating. The upper portion of the nut shall be deformed to provide self-locking feature

4 Push-out and torque-out values shall be as specified in Table II or Table III, as applicable, when properly installed in the following materials:
Steel, alloy, nuts installed in cold rolled steel sheet
Steel, corrosion-resistant, nuts in 2024-T4 aluminum alloy sheet

5 Steel, alloy, nuts installed in 2024-T4 aluminum alloy sheet

6 Military part number:
Consists of the letter M, the basic number of the specification sheet, and a dash number taken from Table I, plus letter L for self-locking nut or leave blank for non-locking feature.

Example of military part number:



M45838/3-1CL Indicates a Nut, Plain, Clinch , Self locking, (Self clinching, Knurled collar, 450°F)
0.0860-56 UNJC-3B internal thread, 303 CRES, cleaned, descaled and passivated, solid film lubricant coating

TABLE II
For dash numbers with no (e)

SIZE	APPROX INSTALL FORCE lbf	PUSH-OUT, lbf MIN	TORQUE-OUT, lbf in MIN
0 0860	2600	160	20
0 1120	2700	160	20
0 1380	2700	180	40
0 1640	4000	360	60
0 1900	4000	360	60
0 2500	5000	550	120
0 3125	7000	800	225
0 3750	8200	1000	325

TABLE III
For dash numbers with an (e)

SIZE	APPROX INSTALL FORCE lbf	PUSH-OUT, lbf MIN	TORQUE-OUT, lbf in MIN
0 0860	2200	80	10
0 1120	2400	120	20
0 1380	2600	160	30
0 1640	3400	240	45

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Recommended Installation Procedures:

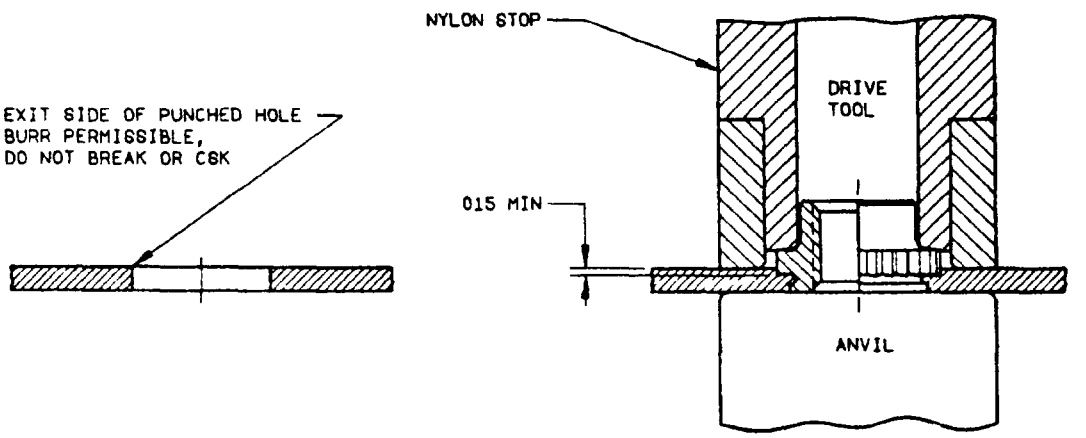
a. Drill or punch hole (Corrosion-resistant steel shall be drilled).

b. Locate pilot squarely in hole. Use exit side of punched hole.

c Using a drive tool and anvil (as shown below), apply sufficient force to cause a .015 inch minimum penetration into the structure for a full 360°. Tool shown is a depth-stop type tool which installs nut to proper depth when nylon stop bottoms against sheet surface Tool mandatory from Rosan Products, Santa Ana, CA - CAGE CODE 83324

(B)

d Install screw or bolt from pilot side of nut



Custodians:
Army - AR
Navy - AS

Review activities
Army - AV, MI
DLA - IS

User activities:
Army - AT, CR, ER, GL, ME
Navy - AS, MC, SH

Preparing activity
Army - AR

(Project 5310-1675)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-N-45938/3B	

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last First MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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